

NEURAL BELLMAN OPERATORS

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We develop Neural Bellman Operators for continuous-time economic control with recursive utility, constraints, and endogenous preferences. The computation separates policy evaluation from feasible improvement and returns an economic error account. Verification bounds translate evaluation, action, and boundary errors into value and welfare bounds; a positive-weight implementation supplies viscosity selection. We train multilayer critics and actors on a nonhomothetic consumption problem and compute full-domain interval certificates, independently of reference-grid error. Matched direct-HJB and joint-loss comparisons identify both successful solves and consequential optimization failures. Further computations train neural preference-adjustment policies, test dense nonquadratic capital economies and stochastic diffusion traces, and solve finite dynamic Cournot games with independent best responses. Action-domain sensitivity and out-of-domain payoff diagnostics reveal which conclusions transfer beyond a numerical approximation. The framework retains separate criteria for economic optimality, equilibrium deviations, and numerical stationarity.

KEYWORDS: Dynamic programming, Neural approximation, Recursive utility, Endogenous preferences, Viscosity solutions, Dynamic games.

1. INTRODUCTION

Quantitative economic models increasingly place several margins of adjustment inside the same dynamic decision problem. An investor chooses consumption and risk exposure while managing a preference state. A sophisticated consumer chooses today knowing that tomorrow's self will reconsider. A firm invests while anticipating the investment policies of its rivals. These problems share a computational structure: a continuation value must correctly evaluate a policy, and that policy must satisfy the appropriate optimality or equilibrium condition. The two requirements

are distinct. A perfectly evaluated policy need not be optimal, and an attractive policy evaluated by an inaccurate continuation value need not improve welfare.

This paper develops Neural Bellman Operators (NBO) around this distinction. A critic represents continuation utility. A feasible actor represents the controls that enter the economic generator. The critic is fitted to a policy-evaluation equation, including its terminal and boundary data. Holding the critic and its derivative channels fixed, the actor improves its own Hamiltonian. An independent maximization step measures the remaining feasible-action gap. The resulting error decomposition concerns economic values and deviations, rather than only parameter gradients or the scalar quantity minimized by an optimizer.

The motivation remains continuous-time economic control, including recursive utility and costly preference formation in the motivating model of [Qi \(2025\)](#). We do not replace these problems by a prediction task. Instead, we make explicit how a neural approximation is connected to the original decision problem. In the smooth case, the connection follows from a verification bound. In the nonsmooth case, it follows from a monotone Bellman approximation and a comparison principle. The actor and critic may be trained by different numerical procedures in either case. Their economic interpretation is determined by the equations they satisfy and by the errors that are verified after training.

The paper makes three contributions. First, it specifies a neural evaluation–improvement map whose derivative graph, feasible action class, economic boundary data, and verification output are explicit. Second, it connects differential and positive-weight implementations through economic error accounts, including a computable continuous-domain certificate for a trained nonlinear consumption problem. Third, it implements this distinction across endogenous preferences, coupled capital economies, and strategic agents, while retaining recursive utility and temporal selves within the same economic formulation. The numerical comparisons identify approximation costs and failures rather than attributing optimality to parameter stationarity. Monotonicity, comparison, and approximate policy iteration are inherited mathematical ingredients; the implemented economic specification and verified numerical chain are the objects studied here.

The numerical program now includes generic multilayer neural fits as well as the analytical laboratories. A stopped consumption economy has independent upwind references and full continuous-domain neural certificates. Preference adjustment has a same-model neural–Bellman comparison and an action-domain sensitivity study. Dense nonlinear capital models exercise automatic differentiation, stochastic trace estimation, and state coverage. Dynamic Cournot profiles receive independent fixed-rival best-response calculations. Homothetic portfolios, temporal-self formulas, the nonsmooth exit problem, and the exact coupled quadratic family remain as distinct economic and algebraic checks. No analytical or matrix calculation is counted as a generic neural training run.

1.1. *Relation to the literature*

Neural residual methods build on numerical dynamic programming and function approximation, not on a new principle of optimality. Projection and related methods exploit economic structure to approximate a Bellman equation; see [Judd \(1998\)](#). Derivative approximation by smooth neural networks provides useful representation results ([Hornik et al., 1990](#)), but does not establish that a particular finite architecture or optimizer solves a nonlinear boundary-value problem.

The differential critic is closely related to the neural PDE approach in [Sirignano and Spiliopoulos \(2018\)](#). Their DGM formulation includes differential and boundary conditions and treats HJB equations. The deep BSDE method of [Han et al. \(2018\)](#) also includes an HJB application. The distinguishing object here is the combination of an explicit feasible actor, block-specific evaluation, and economic error accounting across control and equilibrium problems. An analytical formula for the maximizing action is helpful when available, but is not a universal dividing line between neural PDE methods. Likewise, recursive dependence on the value is a nonlinear zeroth-order term; full nonlinearity in the PDE sense concerns the highest derivatives.

Recursive utility separates intertemporal substitution from risk aversion ([Epstein and Zin, 1989](#), [Duffie and Epstein, 1992](#)). Its infinite-horizon formulation requires particular care about utility-process selection and transversality; [Herdegen et al. \(2021\)](#) explains why a stationary algebraic solution alone is insufficient. The time-

inconsistency extension is an equilibrium problem in the sense of Björk et al. (2016), not an ordinary control problem with a relabelled flow payoff. Finally, stochastic approximation results require their own step-size, stability, and limiting-dynamics conditions (Borkar, 2024). They do not convert constant-step Adam into a convergence theorem for economic optimality.

Section 2 specifies the economy. Section 3 defines the implemented map and Section 4 establishes its error accounts. Section 5 completes the differential theorem–algorithm–experiment chain. The subsequent sections retain and extend the portfolio, recursive-utility, preference-adjustment, temporal-self, dynamic-game, and scaling applications. The appendix proves the principal results. The supplement gives arithmetic details, application-specific assumptions, every seed-level comparison, and sensitivity results.

2. THE CONTROLLED ECONOMY

Let W be an m -dimensional Brownian motion on a filtered probability space satisfying the usual conditions. The state $S = (u, X)$ contains internal and external state variables. An action $a = (\theta, c, \pi, \dots)$ combines internal adjustment and external choices. The controlled state evolves as

$$dS_s = b(s, S_s, a_s) ds + \sigma(s, S_s, a_s) dW_s, \quad a_s \in \mathcal{A}(s, S_s). \quad (1)$$

The compact feasible correspondence $\mathcal{A}$ is part of the model. Actions are progressively measurable and must generate a well-defined state process satisfying the declared state restrictions. State viability does not follow from bounded actor outputs alone.

We work initially on a finite horizon T . A computational domain D is either invariant, equipped with a specified reflection mechanism, or stopped on exit. In the stopped case let $\tau = T \wedge \inf\{s \geq t : S_s \notin D\}$. The payoff $g(\tau, S_\tau)$ is specified on the entire parabolic stopping boundary, including the terminal face. For a fixed admissible policy a , utility is the policy-specific solution

$$Y_s^a = g(\tau, S_\tau) + \int_s^\tau f(v, S_v, a_v, Y_v^a) dv - \int_s^\tau Z_v^a dW_v. \quad (2)$$

Adjustment costs are included in f . For example, time-additive utility has $f(s, S, a, y) = r_0(s, S, a) - \rho y$, where r_0 includes all current resource and adjustment costs. Define $V^a(t, x) = Y_t^{a, t, x}$ and $V^*(t, x) = \sup_a V^a(t, x)$. The value inside a candidate policy's recursive objective is its own utility process, not the unknown optimum inserted into every candidate objective.

ASSUMPTION 1—Finite-horizon comparison class: *The controlled state problem is well posed for the admissible policies used below. The coefficients are continuous, locally Lipschitz in the state uniformly over actions, and bounded on the compact stopped domain. The payoff is continuous and bounded. On an invariant utility interval containing all policy values, f is continuous in its arguments and uniformly Lipschitz in utility with constant $L \geq 0$. The admissible class is stable under concatenation and has the measurable-selection properties needed for dynamic programming. The resulting HJB problem and the fixed-policy problems obey comparison in the bounded continuous class with the specified boundary conditions.*

The comparison requirement identifies both a function class and a boundary problem. It is not asserted merely because the primitive coefficients are continuous. On a bounded, regular Dirichlet domain, a boundary barrier is one way to verify attainment of the stopping payoff. Reflection requires the corresponding Neumann or oblique derivative condition and a well-posed reflected state process. Unbounded-state applications require growth bounds and localization. These conditions are checked separately from a neural residual.

For a smooth function v , write

$$\mathcal{L}^a v = b(t, x, a)^\top Dv + \frac{1}{2} \text{tr}\{\sigma(t, x, a)\sigma(t, x, a)^\top D^2 v\}, \quad (3)$$

$$\mathcal{H}^a[v] = f(t, x, a, v) + \mathcal{L}^a v, \quad F[v] = -v_t - \sup_{a \in \mathcal{A}(t, x)} \mathcal{H}^a[v]. \quad (4)$$

The value solves $F[V^*] = 0$ with the specified boundary data, in the viscosity sense where classical derivatives do not exist. The diffusion covariance is the positive semidefinite matrix $\sigma\sigma^\top$. No claim that the product of two arbitrary positive semidefinite matrices is symmetric positive semidefinite is needed.

A continuous martingale representation can also be used when its quadratic variation is Markov in an augmented state. If $d\langle M \rangle_s = Q(s, S_s, a_s)ds$, the covariance in (3) becomes $\sigma Q \sigma^\top$. A merely predictable process Q_s is not, without further state augmentation, a deterministic coefficient of a Markov HJB equation. Quadratic variation by itself also does not imply conditionally Gaussian increments. The implementation in this revision uses the Brownian model (1).

2.1. Infinite horizon and economic normalization

For a stationary problem, the boundary-value equation is $\sup_a \{f(x, a, V) + \mathcal{L}^a V\} = 0$. We use an infinite-horizon result only after specifying admissibility, integrability, and the selected utility process. In the time-additive case, a sufficient verification requirement is that the localized Itô identities are uniformly integrable and $\mathbb{E}[e^{-\rho T} |V(S_T)|] \rightarrow 0$ for the policies being compared. Recursive problems require an appropriate utility comparison and transversality condition instead of simply borrowing the additive one.

Cardinal transformations also require care. Multiplying a fixed utility function by a positive constant does not change its maximizer. Adding a function of an *endogenously controlled preference state* generally does. Consumption units, preference normalization, and terminal utility are therefore primitive economic choices in Section 8.

3. NEURAL BELLMAN OPERATORS

The actor $a_\phi(t, x)$ and critic $v_\xi(t, x)$ represent policies and continuation values. An internal actor and an external actor can be separate parameter blocks; their joint action remains the input to the same economic generator. A smooth activation is useful for exact differentiation, while a nonsmooth neural representation can be used with the positive-weight Bellman formulation below.

3.1. Feasibility and boundary data

For a box $[a, \bar{a}]$, a sigmoid map produces interior feasible actions but does not attain its endpoints at finite weights. Binding constraints can instead be represented by a projection onto the closed interval, by an explicit boundary candidate, or by a

finite action set including the endpoints. The constrained portfolio calculation uses a projection and can attain a zero risky share exactly. On a simplex, a probability map supplies a feasible interior point and boundary faces must again be considered when certifying the optimum. General constraints require a feasible parameterization, a certified projection, or a constrained action solver. A finite penalty is not an exact feasibility certificate.

On a finite horizon, an elementary hard-terminal architecture is

$$v_{\bar{\xi}}(t, x) = G(x) + (T - t)N_{\bar{\xi}}(t, x). \quad (5)$$

This enforces the terminal face, but not automatically a lateral boundary. A compatible lifting G_B and a factor d_B vanishing on all Dirichlet faces give $v_{\bar{\xi}} = G_B + d_B N_{\bar{\xi}}$. Alternatively, boundary residuals are trained separately and retained explicitly in the error bound. A reflected preference state requires its own derivative condition; a terminal architecture does not enforce it. For recursive utility, a domain-preserving transformation must additionally keep the continuation value in the domain of the aggregator.

3.2. The block-specific computational graph

Let ν be a specified distribution of state–time points and let sg denote stop-gradient. A critic step minimizes

$$L_C(\bar{\xi}; \bar{\phi}) = \mathbb{E}_{\nu} \left[-\partial_t v_{\bar{\xi}} - \mathcal{H}^{\text{sg}(a_{\bar{\phi}})}[v_{\bar{\xi}}] \right]^2 + L_B(\bar{\xi}), \quad (6)$$

where L_B is absent only for boundary conditions enforced exactly. The policy, sampling locations, and any rival actions are detached. The value, time derivative, state gradient, and Hessian remain differentiable with respect to the critic parameters.

An actor step minimizes

$$L_A(\phi; \bar{\xi}) = -\mathbb{E}_{\nu} \mathcal{H}^{a_{\phi}}[\text{sg}(v_{\bar{\xi}}), \text{sg}(Dv_{\bar{\xi}}), \text{sg}(D^2v_{\bar{\xi}})]. \quad (7)$$

All critic derivative channels are fixed in this step. There is no critic gradient from L_A and no actor gradient from L_C . A printed sum $L_A + \omega L_C$ may be a reporting

device, but it is not jointly differentiated. The implementation exposes separate functions and regression tests for these derivative paths.

The distinction changes the population problem. With one action, $f(v) = -v$, $T = 1$, and terminal utility one, the correct solution is $v^*(t) = e^{t-1}$. The perturbation $v_\epsilon = v^* - \epsilon(1 - t)$ preserves the terminal condition. The joint objective used in the reviewed draft satisfies

$$\Delta(L_A + L_C) = -\frac{\epsilon}{2} + \frac{7\epsilon^2}{3}, \quad \Delta L_C = \frac{7\epsilon^2}{3}. \quad (8)$$

Thus the revised critic objective has the correct root as its minimizer along this perturbation, whereas the joint objective does not. The same problem is visible in a stationary equation: $-H + \omega H^2$ prefers $H = 1/(2\omega)$, not $H = 0$. No choice of finite positive weight resolves this identification error.

The actor objective is a local policy-improvement surrogate. Under differentiability of the policy value, the derivative of lifetime recursive utility from (t, x) instead involves the adjoint weight

$$D_\phi V^{a_\phi}(t, x) = \mathbb{E}_{t,x} \int_t^\tau e^{\int_t^s f_y(v, S_v, a_v, V^{a_\phi}) dv} D_a \mathcal{H}^{a_\phi}[V^{a_\phi}](s, S_s) D_\phi a_\phi(s, S_s) ds, \quad (9)$$

subject to the differentiability and integrability conditions given in the supplement. Arbitrary domain sampling does not reproduce this occupation and recursive-utility weighting. NBO uses (7) for approximate pointwise improvement and verifies the remaining action gap; it does not identify that loss with an exact lifetime policy gradient.

3.3. Conditioning and stopping

A positive, parameter-independent state weighting can precondition the critic equation without changing its zero set. If that weight depends on learned quantities, its derivative must be included in the stated critic objective. The homothetic laboratory divides the Hamiltonian by its positive coefficient and differentiates that entire normalized critic residual. In the actor step the same coefficient is detached. Error conversion to the original economic equation includes the scale factor.

Algorithm 1 NBO evaluation, improvement, and acceptance

Require: Model, admissible actions, utility domain, boundary data, training and validation domains.

1: Initialize a feasible actor and a boundary-compatible critic.

2: **for** each outer iteration **do**

3: Freeze actor outputs. Update only the critic using (6) or a positive-weight Bellman evaluation residual.

4: Freeze the critic and every derivative channel. Improve only the actor using (7) or a feasible action solver.

5: Evaluate boundary error, policy-evaluation error, and the independent feasible-action gap.

6: Record the initialization, updates, conditioning, unsuccessful attempts, and stopping rule.

7: **if** the declared economic error criterion is met **then**

8: Return the policy, value, verification domain, and error report.

9: **end if**

10: **end for**

11: Return the computed approximation and its measured errors; do not infer optimality from optimizer stationarity.

A zero residual target does not make learning dynamics stationary. For the scalar loss $\ell(z) = a^2 z^2 / 2$, gradient descent is stable only when $0 < \eta a^2 < 2$. Likewise, full differentiation of $\delta^2 / 2$ gives $\delta \nabla \delta$, while a semi-gradient TD update differentiates only the current-value term. The two computations must not be compared using inconsistent signs or discount conventions. If f already includes $-\rho v$, a time discretization must not add a second independent discount to the same continuation equation.

Constant-step Adam is a numerical choice in the calibration laboratory, not an application of a two-timescale theorem. A genuine two-timescale analysis would require nonsummable and square-summable step sizes, an actor-to-critic ratio tending to zero, controlled noise, bounded iterates, and suitable attracting limiting dynamics (Borkar, 2024). A stationary critic gradient need not imply a zero residual,

and a stationary multi-agent gradient need not imply Nash optimality. The results below therefore concern verified approximation errors rather than an assumed universal optimizer convergence property.

3.4. The implemented map and its verification output

Fix the economy, boundary realization, network classes, optimization budget, and verification procedure, denoted collectively by ϑ . Let $\mathcal{P}$ be the admissible feedback policies, Ξ the critic parameter space, and Ω the sampling streams. The implemented NBO is the numerical map

$$\mathcal{N}_\vartheta : \mathcal{P} \times \Xi \times \Omega \longrightarrow \mathcal{V}_\vartheta \times \mathcal{P}_\vartheta \times \mathcal{C}_\vartheta, \quad (a, \xi, \omega) \longmapsto (v, \tilde{a}, C). \quad (10)$$

The first component is approximate policy evaluation; the second is feasible improvement; the third records verification of the resulting pair. A record identifies the model, parameters, domain, boundary faces, evaluation error, maximization error, coverage, discretization errors, and unchecked assumptions. An unavailable bound is not zero. Failure to obtain a useful certificate is a possible output of the map. With its sampling stream fixed, the implementation is deterministic. We do not assert that a network learns an unrestricted Bellman operator on a function space.

The evaluation–improvement structure is an instance of approximate policy iteration, including its partial-evaluation variants (Scherrer et al., 2015). Neural representation does not introduce a new principle of optimality, and monotonicity belongs to the underlying positive-weight scheme, not to gradient descent. The NBO specification joins the block-specific derivative graph to boundary and utility-domain restrictions and to a common economic error account. Sections 5 and 8.4 implement the differential and positive-weight versions of this map, respectively, with independent verification of their outputs.

4. VERIFICATION, APPROXIMATION, AND VISCOSITY SELECTION

For a feasible policy a and a smooth candidate v , define

$$r^a[v] = -v_t - \mathcal{H}^a[v], \quad g^a[v] = \sup_{b \in \mathcal{A}} \mathcal{H}^b[v] - \mathcal{H}^a[v] \geq 0. \quad (11)$$

The first error concerns evaluation; the second concerns improvement. Their difference is the maximized HJB residual: $F[v] = r^a[v] - g^a[v]$. We keep the two quantities separate because the policy value and its regret depend differently on them.

THEOREM 1—Finite-horizon economic error bound: *Under Assumption 1, suppose a bounded smooth candidate v in the declared utility interval and an admissible Markov policy a satisfy $|r^a[v]| \leq e$, $0 \leq g^a[v] \leq \eta$ throughout the verification domain. Suppose the boundary payoff error is at most B . Any reflected boundary condition is imposed exactly. Set $A_L(s) = (e^{Ls} - 1)/L$ for $L > 0$ and $A_0(s) = s$. Then, for $s = T - t$,*

$$|v(t, x) - V^a(t, x)| \leq e^{Ls}B + A_L(s)e, \quad (12)$$

$$|v(t, x) - V^*(t, x)| \leq e^{Ls}B + A_L(s)(e + \eta), \quad (13)$$

$$0 \leq V^*(t, x) - V^a(t, x) \leq 2e^{Ls}B + A_L(s)(2e + \eta). \quad (14)$$

Consequently, exact evaluation, exact boundary data, and a zero feasible-action gap identify the optimal value and an optimal policy.

The theorem replaces an assertion about a global minimizer of a composite loss with a statement about the exact computational objects NBO estimates. It allows errors, does not require that a finite network class contain V^* , and bounds performance against all admissible policies covered by comparison. A small sampled residual is not yet the uniform hypothesis of the theorem. Section 4.3 addresses that distinction.

4.1. A positive-weight recursive Bellman approximation

Let P_h^a be a nonnegative transition/interpolation operator on a finite state set, with row sums one after adjoining absorbing states. Boundary values are fixed. For a continuation array w , define $\Psi_h^a w$ pointwise as the scalar solution

$$z = P_h^a w + hf(x, a, z), \quad T_h w = \max_{a \in \mathcal{A}_h(x)} \Psi_h^a w. \quad (15)$$

For stationary calculations assume $f_y \leq -\lambda < 0$ on an invariant value domain, and require that all scalar roots remain in that domain. The map $z \mapsto z - hf(x, a, z)$

is strictly increasing with slope at least $1 + \lambda h$. Hence the root is unique and Ψ_h^a is order preserving and a sup-norm contraction with factor $q_h = (1 + \lambda h)^{-1}$. The maximum over feasible actions preserves these properties. A root bracket must contain the root; arbitrary clipping of utility values is not part of the definition.

For a finite horizon, the same implicit construction is used backward from the prescribed terminal array. Under the Lipschitz condition in Assumption 1, $hL < 1$ gives a one-step Lipschitz constant at most $(1 - hL)^{-1}$. Discount contraction is not assumed for a recursive parameter regime where it has not been verified. For a time-additive problem, the positive-weight backup $hr_0 + e^{-\rho h} P_h^a w$ is also consistent and monotone. It is the backup used by the NDU reference laboratory.

THEOREM 2—Discrete NBO certificate: *Suppose (15) is well defined on an invariant domain, with contraction factor $q_h < 1$. Let v_h^* and v_h^a denote the fixed points of T_h and T_h^a , respectively. For any neural value array v and deterministic feasible policy a , set*

$$\delta = \|v - T_h^a v\|_\infty, \quad \epsilon = \|T_h v - T_h^a v\|_\infty.$$

Then

$$\|v - v_h^a\|_\infty \leq \frac{\delta}{1 - q_h}, \quad \|v - v_h^*\|_\infty \leq \frac{\delta + \epsilon}{1 - q_h}, \quad \|v_h^* - v_h^a\|_\infty \leq \frac{2\delta + \epsilon}{1 - q_h}. \quad (16)$$

The certificate covers the stated finite state and action model. Continuous actions require an additional certified maximization or action-coverage error.

This result applies regardless of how the neural array was obtained. Neural projection need not itself be monotone. What matters is that the output satisfies the monotone Bellman equation to the displayed tolerance. The distinction makes it possible to use flexible representations while retaining a comparison-based acceptance test.

THEOREM 3—Viscosity-consistent neural Bellman approximation: *Consider stationary problems satisfying comparison in a bounded continuous class and the strict utility monotonicity used in Theorem 2. Suppose the positive-weight schemes are stable, locally consistent with the HJB operator, and attain the prescribed boundary data in the limit. The*

feasible action approximations become dense with vanishing maximization error. If neural arrays and feasible actors satisfy

$$\delta_h + \epsilon_h = o(h), \quad (17)$$

then their interpolated values converge locally uniformly to the unique viscosity solution. Uniform convergence up to the boundary follows when boundary barriers give uniform boundary attainment. Their discrete policy-value arrays converge to the same limit. No assertion of convergence of the policy functions themselves is required.

Here stability, local consistency, comparison, and boundary attainment are numerical-analysis hypotheses, not consequences of calling a representation neural. The proof is given in the appendix. A finite-horizon version uses backward error accumulation: terminal error tends to zero and the sum of one-step errors tends to zero after multiplication by the one-step Lipschitz factors. This covers finite-horizon recursive problems without imposing stationary discount contraction.

For a diffusion with m Brownian factors, an interior semi-Lagrangian construction uses the $2m$ locations $x + hb \pm \sqrt{mh} \sigma e_j$, each with weight $1/(2m)$. Positive multilinear interpolation preserves monotonicity. For smooth test functions its interpolation contribution is $O(\Delta x^2/h)$; thus $\Delta x^2/h \rightarrow 0$ is a sufficient interior coupling condition. The mean and covariance of the cubature increments match those of the specified Brownian diffusion to first order. Stopping and reflection require consistent boundary transitions. Negative interpolation weights are not permitted in this construction.

4.2. Why an integrated differential residual does not select the value

Consider the exit problem $F(v') = 1 - |v'| = 0$ on $(-1, 1)$ with zero boundary payoff. Its value is $v^*(x) = |x| - 1$. The smooth functions

$$w_\epsilon(x) = \sqrt{1 + \epsilon^2} - \sqrt{x^2 + \epsilon^2}$$

satisfy the same boundary values and have vanishing integrated squared residual, but converge uniformly to $1 - |x|$. At zero the test function one touches the

wrong limit from above and gives $F(0) = 1 > 0$, violating the viscosity subsolution inequality. The defect is not removed by exact Hamiltonian maximization.

On a grid with spacing h , the positive-weight exit backup is

$$(Tv)_i = -h + \max(v_{i-1}, v_{i+1}), \quad v_0 = v_N = 0. \quad (18)$$

It is an undiscounted shortest-exit problem, not an instance of the stationary contraction theorem. Backward propagation of the boundary values gives $v_i = |x_i| - 1$ exactly. At the peak of the wrong candidate, the *scaled* residual $|w_i - (Tw)_i|/h$ is close to two, rather than close to zero. The laboratory therefore distinguishes correct selection from small integrated differential error.

An explicit positive viscosity term is another possible regularization, provided comparison, boundary convergence, and approximation errors for the regularized equations are controlled before taking the viscosity to zero. Network width or spectral bias is not itself a positive viscosity parameter. Moreover, approximation of a C^2 function and its derivatives does not establish approximation of nonexistent classical derivatives at a kink.

4.3. Coverage and independent maximization

A verification distribution is declared independently of the actor's occupation measure. It includes economically relevant counterfactual states and boundary regions, not only states visited by a current policy. If a residual r has a known Lipschitz bound K on a compact domain and a validation set has fill distance d_{fill} , then

$$\|r\|_\infty \leq \max_{x \in \mathcal{V}} |r(x)| + Kd_{\text{fill}}. \quad (19)$$

The same bound can be applied to an action gap if its modulus is controlled. An empirical mean alone does not give this guarantee. If the sampling density is bounded below and the domain has a uniform interior-volume condition, a Lipschitz residual also admits an L^2 -to-sup estimate with exponent $1/(d+2)$, as derived in the supplement. Its dimension dependence is material.

An independent action solver must provide an *upper bound* on the unobserved maximization error. Trying a few alternative actions supplies only a lower bound on

the true gap. Concavity with a duality gap, interval bounds, exhaustive enumeration of a finite action set, or an action net with a known Lipschitz modulus can provide the needed upper bound. An unverified local actor optimizer cannot certify itself.

4.4. Differential and discrete error accounts

For the stationary additive problem, write $f = r_0 - \rho v$, with $\rho > 0$, and let R_h restrict a candidate to the nodes of a positive-weight scheme. On interior nodes set

$$T_h^a w = q_h(P_h^a w + h r_0^a), \quad q_h = (1 + \rho h)^{-1}.$$

Stopping nodes instead use their prescribed payoff.

PROPOSITION 4—Transfer to a Bellman certificate: *Suppose the represented policy is feasible for the discrete action class and*

$$\sup_a \left\| h^{-1}(P_h^a R_h v - R_h v) - R_h \mathcal{L}^a v \right\|_{\infty, \text{int}} \leq \kappa_h.$$

If the differential evaluation residual is at most e and its feasible-action gap at most η , then, on the interior nodes,

$$\|R_h v - T_h^a R_h v\|_{\infty} \leq q_h h(e + \kappa_h), \quad \|T_h R_h v - T_h^a R_h v\|_{\infty} \leq q_h h(\eta + 2\kappa_h).$$

Boundary discrepancies and finite-action coverage errors are additional accounts.

The proof follows by adding and subtracting $R_h \mathcal{L}^a v$; the maximization inequality uses the consistency error twice. The converse is false for arbitrary oscillatory interpolants. For smooth interior functions, local cubature and interpolation estimates can bound κ_h ; they do not justify assigning zero error to a stopping or reflected boundary without checking its realization.

For finite-horizon maps with Lipschitz factors β_t , let δ_t bound evaluation error, ϵ_t bound improvement error, and B_N bound terminal error. Define

$$E_N = B_N, \quad O_N = B_N, \quad R_N = 2B_N, \quad \begin{cases} E_t = \delta_t + \beta_t E_{t+1}, \\ O_t = \delta_t + \epsilon_t + \beta_t O_{t+1}, \\ R_t = 2\delta_t + \epsilon_t + \beta_t R_{t+1}. \end{cases} \quad (20)$$

Backward comparison bounds policy-value error by E_t , optimal-value error by O_t , and policy regret by R_t . These arrays are computed, rather than replaced by a training loss, in the NDU experiment. Bounds at later time indices must use their own accounts, not necessarily the account at time zero. Representation and incomplete optimization enter δ_t ; action coverage enters ϵ_t ; the error of replacing the controlled diffusion by a scheme is separate. If a uniform continuous-to-discrete policy-payoff discrepancy is bounded by D_h , transferring a discrete regret bound adds $2D_h$. Without that estimate, the certificate remains a certificate for the stated discrete economy. In particular, this accounting does not establish the optimizer rate required by Theorem 3.

5. A TRAINED NEURAL BOUNDARY-VALUE PROBLEM

Consider a household with labor income, risky wealth, and a prescribed retirement payoff on exit. Before exit from $D = (0.5, 2.5)$,

$$dX_s = (0.12 + 0.02X_s - c_s) ds + 0.15X_s dW_s, \quad c_s \in [0.02, 0.6]. \quad (21)$$

Discount is $\rho = 0.08$, flow utility is $\log c$, and exit pays $g(x) = \log(0.12 + 0.08x)/0.08$. This is a specified stopped economy, not a claim about an unbounded-wealth problem. Income and the exit option destroy the homothetic reduction used in the portfolio calibration. Bounded coefficients, a compact positive consumption interval, positive diffusion variance on the closed domain, and continuous endpoint data yield the usual stopped-domain comparison problem. A neural approximation must solve its nonlinear differential equation and both boundary conditions.

The critic has two hidden tanh layers of width sixteen and the lifting

$$v_{\xi}(x) = g(\ell) + \frac{g(u) - g(\ell)}{u - \ell}(x - \ell) + (x - \ell)(u - x)N_{\xi}\left(2\frac{x - \ell}{u - \ell} - 1\right).$$

The actor uses the same hidden architecture and a sigmoid into the consumption interval. Here the optimum is not forced to an endpoint by that representation. Independent verification nevertheless maximizes over the *closed* interval: for $p =$

$v'(x),$

$$c^*(p) = \begin{cases} \text{clip}(1/p, 0.02, 0.6), & p > 0, \\ 0.6, & p \leq 0. \end{cases} \quad (22)$$

Training imports the general differential core used for the earlier routing tests. It does not fit reference values, derivatives, or policy labels.

PROPOSITION 5—Continuous-domain economic certificate: *For a stopped discounted additive economy obeying comparison, let $v \in C^2(D) \cap C(\overline{D})$ and let a be admissible. Suppose throughout D that*

$$|\rho v - r_0^a - \mathcal{L}^a v| \leq e, \quad 0 \leq \sup_b (r_0^b + \mathcal{L}^b v) - (r_0^a + \mathcal{L}^a v) \leq \eta, \quad \|v - g\|_{\partial D} \leq B.$$

Then

$$\|v - V^a\|_\infty \leq B + \frac{e}{\rho}, \quad \|v - V^*\|_\infty \leq B + \frac{e + \eta}{\rho}, \quad 0 \leq V^* - V^a \leq 2B + \frac{2e + \eta}{\rho}. \quad (23)$$

Indeed, constant upper and lower shifts of v dominate the evaluation or optimality residual and the stopping data; comparison gives the first two inequalities and their sum gives the third. No reference-grid error is used in this argument.

The executable verifier covers the entire state interval by dyadic subintervals. Outward-rounded propagation through the stored network encloses v, v', v'' , and the actor on every leaf. Equation (22) supplies a global action enclosure. We also use the concavity bound

$$\log c^* - pc^* - (\log c - pc) \leq \frac{1}{2}C^2(1/c - p)^2, \quad C \geq \max\{c, c^*\}, \quad (24)$$

which follows by completing the square in the strong-concavity inequality on the segment joining the two actions. It does not assume an interior maximizer. Taking the better of this bound and a direct interval enclosure avoids dependency inflation. Terminal-free boundary errors are evaluated separately at both endpoints, including rounding in the lifting constants.

The certificate saves every leaf, verifies that their union is exactly $[0.5, 2.5]$, and binds the parameter and arithmetic files by hashes. Exponential and logarithmic

enclosures use range reduction and polynomial remainder bounds rather than an assumed accuracy of a library transcendental. Its mathematical conclusion is conditional on the stated binary64 arithmetic contract; high-precision and derivative tests check the implementation but are not a formal machine proof. The construction does not estimate a global Lipschitz constant from samples.

5.1. *Controlled comparison and failures*

We use seeds 11, 29, and 47, each with 256 prespecified jittered collocation points. Eight outer rounds have a common cap of 800 L-BFGS inner iterations. NBO allocates each round sixty iterations to evaluation and forty to improvement; direct neural HJB uses the same critic and the global action rule in all one hundred iterations. A joint-loss ablation differentiates the sum of squared residual and negative Hamiltonian through both blocks. Line searches use different numbers of closures, so this is a matched iteration cap, not an assertion of equal floating-point work. Closure counts and elapsed times are saved.

An independent positive-rate upwind policy-iteration solver uses continuous-action maximization within each drift region. Reference grids have 320, 640, 1,280, and 2,560 intervals. Value and derivative comparisons use the finest grid; its refinement differences are not substituted for a rigorous discretization bound. The continuous certificate instead checks the neural functions directly. All three NBO and direct-HJB runs attain the declared interval-verification thresholds. The joint-loss ablation does not: a residual near one half is accompanied by large value and action errors. Width, depth, and update-allocation variations are retained in the supplement, including unfavorable outcomes. The direct method is a strong comparator: these results do not establish that carrying an actor is uniformly faster or more accurate. They establish an end-to-end neural solve with an independently computable welfare bound, and identify a concrete failure of joint differentiation. Fitting and full-domain verification are separately timed because their computational burdens are different.

TABLE I

MULTILAYER CONSUMPTION: APPROXIMATION AND CONTINUOUS-DOMAIN VERIFICATION

Method	Seed	$\ v - V_h^*\ _\infty$	Sampled e	Uniform η	Regret bound
NBO	11	3.30×10^{-4}	6.18×10^{-4}	4.24×10^{-6}	0.0501
NBO	29	5.75×10^{-4}	0.0015	2.66×10^{-6}	0.0500
NBO	47	5.46×10^{-4}	0.0014	4.36×10^{-6}	0.0501
Direct HJB	11	4.14×10^{-4}	9.72×10^{-4}	0	0.0500
Direct HJB	29	3.55×10^{-4}	7.13×10^{-4}	0	0.0500
Direct HJB	47	6.00×10^{-4}	0.0017	0	0.0500
Joint loss	11	5.4142	0.5002	—	—
Joint loss	29	5.4435	0.5003	—	—
Joint loss	47	5.5677	0.5003	—	—

The comparison grid has 2,560 intervals on $[.5, 2.5]$. The last two columns cover the entire continuous interval and consumption set $[.02, .6]$. All six certificates have uniform evaluation error at most .002 and endpoint error below 10^{-12} . Dashes indicate no accepted certificate, not zero error. Reference error and the sampled differential residual are not used as uniform bounds.

6. ANALYTICAL AND NONSMOOTH BENCHMARKS

The time-additive Merton model has wealth dynamics

$$dX = [(r + p(\mu - r))X - c]dt + p\sigma X dW, \quad U(c) = \frac{c^{1-\gamma}}{1-\gamma}.$$

For the stationary infinite-horizon benchmark, set $c = mX$, $V = KX^{1-\gamma}/(1-\gamma)$, and impose the usual admissibility and transversality requirements. Writing

$$p^* = \frac{\mu - r}{\gamma\sigma^2}, \quad R_* = r + p^*(\mu - r) - \frac{1}{2}\gamma(p^*)^2\sigma^2,$$

substitution and maximization give

$$m^* = \frac{\rho + (\gamma - 1)R_*}{\gamma}, \quad K = (m^*)^{-\gamma}. \quad (25)$$

With $\rho = .04$, $\gamma = 2$, $\mu = .08$, $r = .02$, and $\sigma = .2$, these numbers are $p^* = .75$ and $m^* = .04125$. These are stationary constants; an arbitrary finite-horizon problem with an unspecified bequest does not have the same constant consumption rule.

A finite-horizon verification test can use the matching bequest $KX^{1-\gamma}/(1-\gamma)$, explicitly declared rather than silently omitted.

For $\mu = .01$ with no short sales, the unconstrained share is $-.125$ and the constrained optimum is zero at every positive wealth. Its effective return is r and $m^* = .03$. The maximized HJB remains an equality. The action KKT condition is an inequality at the boundary, but it does not turn this regular-control problem into an obstacle problem or create a wealth-dependent kink. The exit experiment in Section 4, rather than this homothetic corner solution, supplies the nonsmooth selection test.

6.1. Executed calibration and interpretation

The replication program trains a homothetic neural subfamily: a positive value coefficient, a sigmoid consumption-share output, and a projected portfolio-share output. These are neural output parameters in a known invariant representation, not an unrestricted multilayer approximation. The two Adam optimizers have learning rates $.025$ and $.008$; each outer iteration contains three critic updates and one actor update. There are 1,600 outer iterations and seeds 0, 1, 2. All attempted runs, including their coefficient and policy traces, are saved. These calibrations optimize the stationary coefficient equation and then check the resulting coefficient against the matching bequest; the coefficient error is a boundary error in that finite-horizon interpretation, not a boundary condition hard-imposed during these particular updates.

Table II reports quantities generated by the new code. The normalized policy-evaluation residual is below 4×10^{-16} in these runs. These results establish that the revised graph and formulas agree in the exactly representable laboratory. They do not establish a broad advantage over a direct analytical calculation. In the exit laboratory, grids with 40, 80, 160, and 320 intervals recover the grid value to within 3×10^{-15} ; the wrong smooth sequence has scaled residual between 1.984 and 1.999. All grid nodes, not a selected interior sample, enter that diagnostic.

TABLE II
CORRECTED PORTFOLIO CALIBRATION: ANALYTICAL TARGETS AND EXECUTED BLOCK UPDATES

Model	Target m	Target p	Runs	Maximum policy error
Time-additive Merton	.04125	.75000	3	5.6×10^{-16}
No-short-sale corner	.03000	.00000	3	2.3×10^{-16}
Recursive matching-bequest test	.04550	.30000	3	7.0×10^{-17}

Policy error is the maximum absolute error in the two constant policy outputs, maximized over all three seeds. Calculations use CPU float64 in the declared homothetic subfamily. Machine-level calibration accuracy is not evidence about an unrestricted network, a GPU runtime, or a general recursive infinite-horizon problem.

7. RECURSIVE UTILITY

Let $a = 1 - 1/\psi$, $b = 1 - \gamma$, and $Y = bV > 0$, with $\psi \neq 1$ and $\gamma \neq 1$. We use the difference-form aggregator

$$f(c, V) = \frac{\rho}{a} \left[c^a (bV)^{1-a/b} - bV \right]. \quad (26)$$

Its consumption derivative is

$$f_c = \rho c^{a-1} (bV)^{1-a/b} > 0. \quad (27)$$

When $\gamma = 1/\psi$, one has $a = b$ and $f = \rho(c^b/b - V)$, a positively normalized time-additive CRRA aggregator. The factor b must not multiply the consumption term outside its power. Doing so reverses consumption monotonicity when $\gamma > 1$. This distinction follows the difference-form convention discussed by [Herdegen et al. \(2021\)](#).

For a homothetic candidate $V = KX^b/b$, the maximization equation becomes

$$0 = \sup_{m,p} \left\{ \frac{\rho}{a} [m^a K^{-a/b} - 1] + r + p(\mu - r) - m - \frac{1}{2} \gamma p^2 \sigma^2 \right\}. \quad (28)$$

The consumption first-order condition is $\rho m^{a-1} K^{-a/b} = 1$. Substitution yields

$$p^* = \frac{\mu - r}{\gamma \sigma^2}, \quad m^* = \psi \rho + (1 - \psi) \left[r + \frac{(\mu - r)^2}{2\gamma \sigma^2} \right], \quad K = [\rho (m^*)^{a-1}]^{b/a}. \quad (29)$$

For the retained parameters $\gamma = 5$, $\psi = 1.5$, $\rho = .04$, $\mu = .08$, $r = .02$, $\sigma = .2$, this gives $(m^*, p^*) = (.0455, .3)$.

7.1. A well-defined validation problem at the retained parameters

The numerical calibration uses (29) as a finite-horizon matching-bequest test, not as an unqualified infinite-horizon theorem. Choose $T > 0$, stop wealth on leaving $[\underline{x}, \bar{x}]$ with $0 < \underline{x} < \bar{x} < \infty$, and impose the candidate Kx^b/b on the stopping boundary and at T . Consumption shares are in a compact positive interval containing .0455, and portfolio shares are in a compact interval containing .3.

For these retained parameters, $a > 0$ and $b < 0$. Since c ranges over a compact positive interval, there exist $0 < \underline{Y} < \bar{Y} < \infty$ such that $f(c, \underline{Y}/b) \leq 0$ and $f(c, \bar{Y}/b) \geq 0$ for every feasible c . Indeed, $c^a Y^{-a/b}$ tends to zero as $Y \downarrow 0$ and to infinity as $Y \uparrow \infty$. Enlarge the interval so that it contains the boundary candidate. The constants $\bar{Y}/b$ and $\underline{Y}/b$ are respectively lower and upper barriers for the backward utility equation. Scalar BSDE comparison therefore keeps every policy utility in this interval. The aggregator is Lipschitz there. The smooth homothetic candidate maximizes the Hamiltonian and matches all stopping payoffs; Theorem 1 identifies it as the value of this specified finite-horizon problem.

The choice of a matching bequest is an analytical validation device. It does not establish that the same candidate is selected by every infinite-horizon recursive formulation. A separate infinite-horizon application must check its utility-process class and transversality, particularly when risk aversion and intertemporal-complementarity parameters straddle one (Herdegen et al., 2021). The paper retains that economic application, but does not use a scalar root to bypass its selection issue.

A critic can enforce $bV > 0$ through $bv_{\xi} = \exp(N_{\xi})$. With a positive terminal target, logarithmic interpolation between bG and the neural output enforces both the terminal condition and the power domain. Additional barriers or bounded transformations enforce a declared invariant interval. Such transformations are stated in the architecture, not imposed retrospectively after a failed evaluation.

7.2. Stability and baseline design

The recursive experiment in Table II tests the corrected sign, coefficient, and block graph. It does not compare independent DDPG training trajectories. The reviewed manuscript's constructed oscillating curve and manually entered coordinates remain in the archival source, but have no evidentiary role in this revision. A matched neural comparison must use the same utility process, domain, bequest, action restrictions, training budget, and held-out evaluation quantities. It must report all failed seeds and value-domain violations. Neither a zero-centered target nor three successful homothetic calibrations establishes general relative stability.

8. ENDOGENOUS PREFERENCES AND PORTFOLIO CHOICE

The endogenous-preference application retains its internal adjustment margin. Let u be risk aversion, X wealth, and θ adjustment effort. On a declared interval $[\underline{u}, \bar{u}] \subset (1, \infty)$, specify the reflected process

$$du = \theta dt + \sigma_u dW^u + dL^{\underline{u}} - dL^{\bar{u}}, \quad (30)$$

$$dX = [(r + p(\mu - r))X - c]dt + p\sigma_X X dW^X, \quad d\langle W^u, W^X \rangle_t = \varrho dt. \quad (31)$$

The nondecreasing local-time regulators act only at their respective boundaries. This supplies the state mechanism that a bounded adjustment actor does not supply. The numerical reference uses [1.2, 3], so the utility singularity at $u = 1$ is never reached. In an alternative positive-state model one could specify a multiplicative or transformed diffusion, but that would be a different primitive and must be identified as such.

8.1. Cardinal utility and the objective

Fix a reference consumption unit $\bar{c} > 0$. The baseline flow utility is

$$U(c, u) = \frac{(c/\bar{c})^{1-u}}{1-u}, \quad r_0(c, u, \theta) = U(c, u) - \frac{1}{2}k\theta^2. \quad (32)$$

The reference calculation takes $\bar{c} = 1$, preserving the unnormalized CRRA family of the original application. The agent maximizes

$$\mathbb{E}_{t,u,X} \left[\int_t^\tau e^{-\rho(s-t)} r_0(c_s, u_s, \theta_s) ds + e^{-\rho(\tau-t)} G(u_\tau, X_\tau) \right]. \quad (33)$$

Both the payoff and the adjustment cost have the same time-relative discount convention. The terminal/exit payoff and the state domain are stated for each calculation.

A useful family for normalization sensitivity is

$$U_\kappa(c, u) = \frac{(c/\bar{c})^{1-u} - 1}{1-u} + \kappa(u).$$

The baseline corresponds to $\kappa(u) = 1/(1-u)$. The log-continuous normalization corresponds to $\kappa = 0$ and is an economically different preference-formation model. A calibration of κ fixes the cardinal value of a reference consumption stream at each preference state. The choice is not identified by risk-aversion curvature alone.

In particular,

$$U_u(c, u) = \frac{(c/\bar{c})^{1-u} \{1 - (1-u) \log(c/\bar{c})\}}{(1-u)^2}. \quad (34)$$

For $c/\bar{c} > 1$ and $u > 1$ this derivative is positive. At $c = 2$, $u = 2$, $\bar{c} = 1$, it is approximately .84657. This is a derivative of flow utility, not a sign theorem for the endogenous value derivative V_u . The agent's full incentives also depend on future consumption, boundary behavior, and the adjustment technology.

8.2. Hamiltonian, constrained policies, and economic implications

The HJB equation is

$$\begin{aligned} -V_t = \sup_{c,p,\theta} \{ & U(c, u) - \frac{1}{2} k \theta^2 - \rho V + \theta V_u + [(r + p(\mu - r))X - c] V_X \\ & + \frac{1}{2} \sigma_u^2 V_{uu} + \frac{1}{2} p^2 \sigma_X^2 X^2 V_{XX} + \varrho \sigma_u \sigma_X p X V_{uX} \}. \end{aligned} \quad (35)$$

Reflection gives $V_u = 0$ at the two preference boundaries, in the appropriate boundary-value sense. The correlation term is retained with its correct coefficient.

For an interior optimum with $V_X > 0$ and $V_{XX} < 0$, the first-order conditions are

$$c^* = \bar{c} [\bar{c} V_X]^{-1/u}, \quad (36)$$

$$\theta^* = V_u / k, \quad (37)$$

$$p^* = -\frac{V_X(\mu - r)}{X V_{XX} \sigma_X^2} - \frac{\varrho \sigma_u V_{uX}}{X V_{XX} \sigma_X}. \quad (38)$$

For independent box constraints, strict concavity in the corresponding action permits projection of these unconstrained maximizers onto their intervals. If V_{XX} has the wrong sign or constraints couple the actions, a joint feasible maximization is required instead. The formula for p must not be used where its denominator vanishes.

Equation (37) gives a conditional shadow-price interpretation of internal adjustment. Holding V_u fixed, a larger k reduces its magnitude. Across economies, however, V_u itself depends on k , so this is not a global policy comparative static. A valid unconditional implication is that the optimized value is weakly decreasing in k : every fixed admissible policy loses $\frac{1}{2}(k_2 - k_1)\mathbb{E} \int e^{-\rho s} \theta_s^2 ds$ when the cost rises. When an envelope derivative exists,

$$\partial_k V^k = -\frac{1}{2} \mathbb{E}^{a^k} \int_t^\tau e^{-\rho(s-t)} \theta_s^2 ds \leq 0. \quad (39)$$

The sign of the hedging component in (38) remains conditional on V_{uX} and the curvature of wealth value. A negative correlation alone does not determine it.

8.3. Independent two-state reference and identification

The executable reference solves the finite-horizon problem with $T = 1$, $u \in [1.2, 3]$, wealth stopped at $[.5, 2]$, and $G(u, X) = X^{1-u} / (1 - u)$ at exit and terminal time. Parameters are $(r, \mu, \sigma_X, \sigma_u, \varrho, \rho) = (.02, .08, .2, .08, -.3, .04)$. The finite action set is the product of $c/X \in \{.02, .05, .1\}$, $p \in \{0, .375, .75\}$, and $\theta \in \{-.15, 0, .15\}$. A four-point Brownian cubature with positive interpolation weights performs backward induction. The wealth boundary is imposed at every time step; preference overshoots are reflected. This reference defines a finite-action numerical approximation, not continuous-action optimality.

At $(u, X) = (2.1, 1)$, the original 17×25 state grid and 20 time steps give values -11.915211 and -11.956983 for $k = 1$ and $k = 5$. The 25×37 grid and 40 time steps give -11.866768 and -11.908738 . Both select $(c/X, p, \theta) = (.1, .75, -.15)$ at that point. The approximately .048 refinement change is reported separately from training error. The higher-cost value is weakly lower at every recorded node and time. These original references remain useful consistency checks; Section 8.4 supplies actual neural fits and shows how strongly the finite action domain affects their economic comparison.

The model also specifies what a structural estimation exercise would require. For observations $Y_j = \mathcal{M}(S_{t_j}, a_{t_j}; \theta) + \varepsilon_j$, let $m(\theta)$ denote selected population moments of the induced observation process. Local identification requires a rank condition on $D_\theta m$, after accounting for latent preference states, normalization parameters, and observation noise. Variation of a simulated policy with k alone is insufficient. Numerical value and policy errors must additionally be propagated into moment errors. The endogenous-preference application is therefore retained as an economic model with an explicit internal margin, not used as evidence that its structural parameters have already been identified from data.

8.4. Neural solution and action-domain sensitivity

We now apply multilayer NBO to the same two-state economy, with the original cardinal utility, correlation, reflection rule, and stopping payoff unchanged. At each backward time level a two-hidden-layer, width-thirty-two tanh critic approximates its own Bellman continuation; a multilayer categorical actor selects among feasible actions. The reference solution is computed separately and is not used as a training target. The actor is fitted before policy evaluation at that time level, with the next-period critic fixed. The direct comparator eliminates the actor by enumeration of the same action set. Both methods have a cap of 160 L-BFGS inner iterations per time level, divided 100/60 between critic and actor for NBO.

An initial unweighted classification implementation produced a large policy loss. Absorbing wealth-face labels, for which every action has identical value, contaminated the policy fit; classification error also ignored the economic cost of

TABLE III

NEURAL PREFERENCE ADJUSTMENT ON THE SAME FINITE ECONOMY

Method	Seed	Value error	Policy regret	Bound R_0	Midpoint RMS
NBO	11	0.1679	0.1656	3.3954	0.0438
NBO	29	0.1495	0.1487	2.8503	0.0413
NBO	47	0.1572	0.1014	3.0605	0.0445
Direct Bellman	11	0.1162	0.0261	1.7944	0.0477
Direct Bellman	29	0.1061	0.0261	1.6930	0.0400
Direct Bellman	47	0.1342	0.0269	1.8224	0.0424

The first three error columns use every state and time of the $17 \times 25 \times 21$ grid with 175 actions; R_0 is the time-zero certificate, while the first two columns maximize their independently computed errors over all times. Midpoints compare the time-zero neural critic with the finite-reference interpolant. Continuous-action and controlled-diffusion discretization errors are not included. The accuracy targets were value error .2 and policy regret .1.

a mistaken low-consumption action. The retained implementation excludes those faces from actor training and minimizes expected normalized Bellman action loss, with a small cross-entropy term. The pilot source, weights, and failed output are preserved. This change repairs an implementation failure rather than redefining the economic objective: the deployed action is still evaluated against *all* finite feasible alternatives.

For the principal comparison, $(n_u, n_x, N_t) = (17, 25, 20)$ and $k = 1$. The expanded action set contains 175 triples, with $c/X \in \{.02, .05, .10, .20, .30\}$, $p \in \{0, .375, .75, 1.125, 1.5\}$, and $\theta \in \{-.45, -.30, -.15, 0, .15, .30, .45\}$. Every time-state-action combination enters the error accounts in (20). Independently solving the fixed learned policy gives its actual regret relative to the finite reference. Midpoint value errors and derivative comparisons are additional diagnostics, not continuous-domain certificates. The three-seed record contains a failed NBO accuracy target and reports it without exclusion. The direct comparator is more accurate at this budget. A categorical actor therefore has a measurable approximation cost even when the critic class is identical. Neural representation alone is not a reason to

replace an inexpensive two-state grid solver; the grid here supplies an unusually complete audit of the neural computation.

Action bounds are part of the economy, not innocuous numerical constants. The sensitivity study reproduces the referee’s action-set comparison and extends it to 1,053 actions in the same expanded box, a wider consumption–portfolio–adjustment box, a wider wealth domain, and two finer space–time grids, for both $k = 1$ and $k = 5$. The table in the supplement records lower- and upper-bound frequencies for every control. Consumption and portfolio bounds remain frequently active after the first expansion. Thus state-grid convergence of the old 27-action model does not establish convergence to an unconstrained control optimum. The wider sets describe different constrained economies; refinement within a fixed box is a different comparison. No continuous-action coverage or controlled-diffusion discretization error is set to zero in the finite certificate. This preserves the economic application while making its constraint dependence observable.

9. SOPHISTICATED TEMPORAL SELVES

To define present bias in continuous time, fix the discount kernel

$$h(s) = \beta e^{-\rho s} + (1 - \beta)e^{-(\rho+\kappa)s}, \quad 0 < \beta \leq 1, \quad \kappa > 0. \quad (40)$$

It satisfies $h(0) = 1$ and discounts the immediate future more heavily than the distant future when $\beta < 1$. It is a finite-duration continuous-time specification; it is not the assertion that multiplying current utility by β creates present bias.

For a candidate Markov policy a^* , define the two continuation functions

$$W_j(t, x) = \mathbb{E}_{t,x}^{a^*} \left[\int_t^T e^{-\rho_j(s-t)} u(a_s^*) ds + e^{-\rho_j(T-t)} G(S_T) \right], \quad (\rho_1, \rho_2) = (\rho, \rho + \kappa). \quad (41)$$

Write $V = \beta W_1 + (1 - \beta)W_2$. Each component evaluates the *same* future policy:

$$0 = (W_j)_t + \mathcal{L}^{a^*} W_j + u(a^*) - \rho_j W_j, \quad W_j(T) = G. \quad (42)$$

The current self’s spike-deviation condition is

$$a^*(t, x) \in \arg \max_{a \in \mathcal{A}(t, x)} \{u(a) + \mathcal{L}^a V(t, x)\}. \quad (43)$$

The deviations change both current flow utility and the transition of the continuation state. They do not hold all dynamic effects at the equilibrium action. The appendix derives (43) from a deviation on $[t, t + \epsilon]$ followed by a^* , thereby fixing the equilibrium concept.

A temporal-self NBO uses two policy-evaluation critics for (42) and one actor for (43). The current actor does not optimize the sum of future selves' parameter objectives. A target actor can stabilize a numerical iteration, but its agreement with the current actor is not an equilibrium certificate. The relevant diagnostics are component evaluation errors and one-shot deviation gains. The benchmark $\beta = 1$ collapses to the usual exponentially discounted problem.

9.1. *An equilibrium calculation with endogenous continuation values*

For deterministic wealth $dX = (rX - c)dt$, CRRA utility with $\gamma > 1$, and an infinite horizon, consider $c = mX$. Set $b = 1 - \gamma$ and $\lambda_j(m) = \rho_j - b(r - m) > 0$. Then

$$W_j(X) = \frac{m^b}{\lambda_j(m)} \frac{X^b}{b}, \quad 1 = m \left[\frac{\beta}{\lambda_1(m)} + \frac{1 - \beta}{\lambda_2(m)} \right]. \quad (44)$$

The expression on the right is strictly increasing in m on its admissible interval, starts at zero, and diverges at the first pole for $\beta > 0$. The equilibrium root is unique in this homothetic class. Its derivative with respect to β is negative because $1/\lambda_1 > 1/\lambda_2$. Thus stronger present bias raises consumption in this specified equilibrium, not merely at a fixed shadow price.

With $r = .02$, $\rho = .04$, $\gamma = 2$, and $\kappa = .2$, the roots for $\beta = 1, .9, .7, .5$ are respectively .030000, .031372, .034606, and .038760. The code independently optimizes finite-duration deviations and records the gain divided by duration as the duration shrinks. For $\beta = .7$, this quantity declines from approximately 4.60×10^{-5} at duration .1 to 4.66×10^{-9} at duration .001. This is a diagnostic of the derived spike equilibrium. It does not assert that every finite-duration commitment deviation is unprofitable, which would be a different solution concept.

10. DYNAMIC GAMES

The dynamic Cournot application retains firm-specific capital and investment.

Firm i has

$$dK^i = (I^i - \delta K^i)dt + \sigma_K K^i dW^i, \quad q_i = AK^i,$$

and payoff rate

$$r_i = P_N(K)q_i - I^i - \frac{1}{2}b_i(I^i)^2, \quad P_N(K) = \max \left\{ 0, P_0 - \frac{1}{M_N} \sum_j q_j \right\}.$$

For each experiment, the market-size normalization M_N , capital domain, investment constraints, covariance, terminal/liquidation values, and horizon are part of the primitives. Keeping $M_N = 1$ is a fixed-market comparison. Taking $M_N = N$ is a growing-market comparison. They do not have the same competitive-limit interpretation.

Given rivals' Markov policies a^{-i} , a player's best-response equation maximizes only its own action. In the stationary additive case,

$$\rho_i V_i = \sup_{a_i \in \mathcal{A}_i} \{r_i(x, a_i, a^{-i}(x)) + \mathcal{L}^{a_i, a^{-i}} V_i\}. \quad (45)$$

A Markov-perfect equilibrium is a feasible profile satisfying these simultaneous unilateral conditions, with the same boundary and admissibility conventions. A zero policy-evaluation residual can hold for any consistently valued profile and therefore does not identify this equilibrium.

10.1. Player-specific differentiation and deviation bounds

Player i 's critic is trained with every policy frozen. Player i 's actor then differentiates only

$$-\mathbb{E}_V \mathcal{H}_i^{a_i(\phi_i), \text{sg}(a^{-i})} [\text{sg}(V_i), \text{sg}(DV_i), \text{sg}(D^2 V_i)]. \quad (46)$$

The updated parameter is ϕ_i , not the full vector of all players' parameters. Rival actions are detached, and player j 's loss never contributes to player i 's actor gradient. An explicit regression test exercises this graph.

For each player, define the unilateral Hamiltonian gap by maximizing $\mathcal{H}_i$ over a_i while keeping rivals fixed. Applying Theorem 1 to that best-response problem bounds its finite-horizon exploitability by $2e^{L_i T} B_i + A_{L_i}(T)(2e_i + \eta_i)$. In a discounted finite-state game, Theorem 2 gives $(2\delta_i + \epsilon_i)/(1 - q_i)$. These are player-specific bounds; a sum of critic losses is not a substitute. They remain valid without uniqueness or convergence of the simultaneous policy-update dynamics, provided the reported profile satisfies the stated errors and best-response comparison hypotheses.

The static Cournot diagnostic makes the distinction visible. With payoffs $q_i(1 - q_i - q_j)$, symmetric Nash output is $1/3$ per firm. Joint-profit maximization gives a symmetric allocation $1/4$. At the latter profile, a firm optimally deviates to $3/8$, earning a gain of $1/64$. A joint objective can therefore be perfectly optimized and still fail a unilateral equilibrium test. This arithmetic is included as a mandatory regression check.

10.2. Symmetry and market scaling

Identical initial weights do not prove equilibrium uniqueness or permutation equivariance. An equivariant actor can be specified using own capital and a permutation-invariant aggregation of rival states. Even then, one must check unilateral deviations rather than infer equilibrium from architectural symmetry. Multiple starting profiles and unsuccessful best-response searches belong in a numerical report.

Under fixed market size, positive constant capital per firm cannot be called the competitive limit while total output tends to infinity and the truncated price tends to zero. Maintaining that capital entails investment and adjustment costs. Under $M_N = N$, positive capital per firm may instead be compatible with a growing market, but it describes a different sequence of economies. The original unsupported capital table remains in the archive. Section 10.3 replaces its evidentiary role with a fully specified finite dynamic game and independent best-response calculations, without asserting convergence of unrestricted neural game training.

TABLE IV

DYNAMIC COURNOT: INDEPENDENT FIXED-RIVAL BEST RESPONSES

Grid	Steps	Actions	M_2	$I_1(0)$	$V_1(0)$	Max. gain
17^2	30	11	1	0	1.3852	0
17^2	30	11	2	0.4800	2.6737	0
25^2	60	17	1	0	1.3905	0
25^2	60	17	2	0.5500	2.6610	0

Center state is $(K_1, K_2) = (.775, .775)$. Horizon is three, capital lies in $[.05, 1.5]^2$, and investment in $[0, .8]$.

The final column maximizes independently computed dynamic unilateral gains over both firms and every finite-model state and time. Zero denotes the displayed floating-point precision; it is not a continuous-game error bound. Six principal cases additionally include a high-index equilibrium tie rule and asymmetric adjustment costs.

10.3. A dynamic capital-game computation

For two firms, we compute the finite-horizon capital game above with $A = 1$, $P_0 = 2$, $\delta = .1$, $\sigma_K = .15$, $\rho = .04$, $T = 3$, investment in $[0, .8]$, and capital in $[.05, 1.5]^2$. Terminal or joint-domain exit pays $.5K_i$. The baseline quadratic adjustment coefficients are $b_1 = b_2 = 1$; an asymmetric experiment uses $b_2 = 1.2$. Independent diffusion cubature and positive interpolation define the finite dynamic game. Investment nets include both bounds.

Backward induction constructs the two continuation-payoff matrices at every state and time. The solver enumerates pure stage equilibria; alternative deterministic tie rules are retained. If no pure equilibrium exists, the solver records this fact and the smallest maximum stage deviation it finds, rather than declaring equilibrium. After fixing the entire computed rival policy, a *separate backward optimization* permits every own action at every state and time. Its value minus the profile payoff is the unilateral exploitability map. A one-period check with a frozen continuation value alone would not perform this test. All six principal finite-game cases have pure equilibria at every stage and zero detected exploitability to the reported arithmetic precision. A deliberately coarse regression case has a missing pure stage equilibrium and is reported as approximate, with time-indexed deviation bounds. The table includes fixed-market $M_2 = 1$ and growing-market $M_2 = 2$ normalizations

as distinct economies; the resulting investment differences do not prove an $N \rightarrow \infty$ limit. Binding-investment frequencies, alternative equilibrium selections, and the asymmetric case are in the raw record. This is an actual dynamic best-response computation, not the static $1/64$ diagnostic. It is a finite-grid game result; a trained neural game and a continuous-state equilibrium bound are different objects and are not inferred from it.

11. SCALABILITY AND THE DIFFUSION TRACE

The original multi-stock consumption-savings specification has additive utility, independent transitions, and no common resource constraint. It decomposes as $V(k) = \sum_j v_j(k_j)$. A separability-aware comparator solves the one-dimensional components, and identical components can be reused. An unreduced sparse grid is not an appropriate sole comparator for that economy.

To test a coupled generator, the new laboratory instead uses

$$dX = (AX + a)dt + \Sigma dW, \quad \sup_a \mathbb{E} \int_0^\infty e^{-\rho t} [-X^\top QX - a^\top a] dt. \quad (47)$$

The drift is a nonnormal dense matrix $A = -.7I + .06Z/\sqrt{d}$, where Z is generated by the fixed model seed $991 + d$. The payoff matrix is $Q = \text{diag}(1, \dots, 2) + .311^\top/d$. The code verifies that $AQ \neq QA$, so the common independent-coordinate reduction of the original benchmark is unavailable. All methods still exploit the declared quadratic structure. A Riccati solver is therefore the appropriate strong reference, rather than a deliberately unreduced general grid.

For $V = -x^\top Px - c_0$ and $a = -Kx$, policy evaluation solves

$$(A - K - \rho I/2)^\top P + P(A - K - \rho I/2) = -(Q + K^\top K), \quad c_0 = \text{tr}(\Sigma \Sigma^\top P)/\rho. \quad (48)$$

Hamiltonian improvement gives $K = P$. A positive semidefinite P is representable by a signed sum of squared linear neural units; the actor is linear. The exact block laboratory uses a Lyapunov solve for this critic and exact actor improvement. It is an exactly representable NBO subfamily and a matrix-identity test, not evidence that generic stochastic neural optimization has the same convergence rate.

Across dimensions 2, 5, 10, 20, 50 and three stabilizing actor initializations, the block iteration takes six steps. The maximum error in P , in operator norm, is below 10^{-13} relative to an independent algebraic Riccati solver. The policy-evaluation coefficient residual is below 10^{-12} . On the unit Euclidean ball, the actor gap is bounded by $\|K - P\|_2^2$, and the residual bound follows from the operator norm of the quadratic coefficient error. The program saves matrices, iteration traces, and CPU wall times for both methods. These very short, single-process timings are diagnostics and do not establish a fixed-accuracy runtime ranking.

11.1. Trained, coupled nonquadratic capital economies

To separate neural scaling from exact quadratic algebra, consider d positive capital stocks and write $Y_i = \log X_i$. Consumption is $c_i = m_i X_i$, with $m_i \in [.02, 2]$, and

$$dY_i = \left[.10 + .15 \tanh((B_d Y)_i) - m_i - \frac{1}{2}(.15^2 + .10^2) \right] dt + .15 dW_i + .10 dW_0, \quad (49)$$

$$r_0(Y, m) = d^{-1} \sum_i (\log m_i + Y_i) - .10 \bar{m}^2, \quad \bar{m} = d^{-1} \sum_i m_i.$$

The dense row-normalized matrix B_d is generated once from a recorded seed. Horizon is one, discount is .04, and terminal utility is $\bar{Y} - .03 d^{-1} \sum_i (Y_i - \bar{Y})^2$. The actual state domain is $\mathbb{R}^d$; no artificial lateral stopping condition is imposed. Dense non-linear productivity, shared adjustment cost, common noise, and terminal dispersion couple the coordinates. Neither the solver nor its comparator calls a Riccati or Lyapunov routine, and no exact solution is built into the network.

Strict concavity gives a global Hamiltonian maximizer at each queried state. For $p_i = \partial_i v$, its components satisfy

$$m_i^* = \text{clip} \left((dp_i + .2 \bar{m}^*)^{-1}, .02, 2 \right),$$

with the upper bound selected when the denominator is nonpositive. The scalar consistency equation for $\bar{m}^*$ is monotone and is solved by bisection. This supplies

TABLE V
ACTUAL TRAINING IN DENSE NONQUADRATIC CAPITAL MODELS

Method	d	Worst RMS	Worst gap	Failures	Median seconds
NBO	2	0.0161	2.42×10^{-4}	0/3	2.6554
Direct HJB	2	0.0161	0	0/3	3.4870
NBO	5	0.0131	0.0020	0/3	4.1518
Direct HJB	5	0.0129	0	0/3	5.0391
NBO	10	0.0188	0.0052	0/3	6.8652
Direct HJB	10	0.0168	0	0/3	7.6988
NBO	20	0.0320	0.0114	2/3	12.8180
Direct HJB	20	0.0170	0	0/3	13.5001

Maxima are across three seeds, each evaluated at 512 independent points in $[0, 1] \times [-.5, .5]^d$. The failure criterion is RMS residual above .025 or queried action gap above .01 after 600 updates. Both methods use the same critic architecture and global action routine. Times include training diagnostics, not the subsequent Monte Carlo evaluations. These are sampled diagnostics, not uniform economic bounds.

the same feasible maximization routine to the direct-HJB comparator and to action-gap measurement; it does not supply the value function.

Both networks have two hidden tanh layers of width thirty-two. The critic enforces the terminal payoff exactly. The main design uses three seeds at $d = 2, 5, 10, 20$, 600 Adam updates, and batches of 128 independently sampled state-time points. Exact automatic differentiation supplies the trace. The held-out set contains 512 fixed independent points in $[0, 1] \times [-.5, .5]^d$. A common diagnostic target is residual root mean square at most .025 and maximum queried action gap at most .01. The recorded first crossing is not necessarily sustained; final failures are also reported. The experiment measures real neural approximation and optimization costs rather than exact quadratic identities. It also shows why a diagnostic target is not a welfare certificate. Simulated policy paths can leave the fitted state box; Euler policy-payoff estimates from the initial state need not agree with a critic that has a small sampled residual inside that box. A follow-up expands the fitting and verification box to $[-1.5, .5]^d$ and uses 1,000 updates at dimensions ten and twenty. Both designs, paired policy-payoff samples, and forty/eighty-step simulation outputs

are retained. Sampling standard errors do not remove time-discretization bias or certify an optimum. Process high-water memory is labeled as such, not interpreted as an isolated per-network allocation. No high-dimensional uniform residual bound is claimed for these sampled quantities.

11.2. *Exact and stochastic trace computation*

Let Σ include any covariance factor and set $B = \Sigma^\top D^2 v \Sigma$. For independent probes with identity covariance,

$$q = \text{tr}(B), \quad \hat{q} = \frac{1}{K} \sum_{k=1}^K z_k^\top B z_k. \quad (50)$$

Gaussian probes have variance $2\|B\|_F^2/K$. If $r = d - q/2$, then

$$\mathbb{E}(d - \hat{q}/2)^2 = r^2 + \frac{1}{4} \text{Var}(\hat{q}). \quad (51)$$

The extra term depends on the critic. Unbiased trace estimation therefore does not give an unbiased squared-residual objective.

Two conditionally independent probe banks repair this particular bias:

$$\hat{L} = (d - \hat{q}^{(1)}/2)(d - \hat{q}^{(2)}/2), \quad \mathbb{E}[\hat{L} \mid t, x, \xi] = r^2. \quad (52)$$

With dominated differentiation, its gradient is also unbiased. The sample product need not be nonnegative and can have substantial variance, so this is an objective identity rather than a guarantee of stable optimization. Exact low-dimensional derivatives remain the independent check. If the actor or deterministic term depends on the same probes, conditional independence must be re-established; the simple identity cannot be reused automatically.

The relevant computational cost depends on the network evaluation and differentiation cost, Brownian rank, covariance factorization, sample count, optimization iterations, and requested accuracy. Forming a dense Hessian involves quadratic storage; Hessian-vector products can avoid it. Neither operation count by itself is a total-complexity theorem. Active control bounds do not universally create gradient discontinuities, and a noisy trace cannot make a correctly projected box action leave its box. Feasibility and value-derivative accuracy are separate diagnostics.

The independent-bank estimator is also exercised during actual training in the coupled ten-state economy. One implementation uses two independent banks of two Gaussian Hessian-vector probes; the matched single-bank ablation squares a four-probe estimate. Exact traces are used only for the common held-out diagnostics. All three seeds, objective signs, optimization histories, and policy-payoff samples are saved. The two training procedures perform similarly at the retained covariance scale; unbiasedness alone does not establish a finite-budget advantage. A separate Monte Carlo test checks both the product objective and its parameter gradient against exact trace expressions with declared tolerances. These calculations validate the estimator's execution without identifying a random training objective with a deterministic certificate.

12. CONCLUSION

Neural Bellman Operators organize computation around two economic conditions: continuation utility must evaluate the chosen policy, and the policy must satisfy a feasible improvement or unilateral equilibrium condition. The revised formulation preserves recursive utility, constraints, endogenous preference adjustment, temporal selves, and strategic interaction within this common architecture. It also makes terminal data, utility domains, and verification regions explicit.

The main results translate these conditions into value, welfare, and deviation bounds. A positive-weight Bellman approximation provides a viscosity-selection mechanism when classical differentiation is insufficient. The neural representation remains flexible; its economic interpretation comes from the equation and the accompanying error test, not from the printed target or an optimizer's stationary point.

The computations now connect a genuinely trained multilayer NBO to a continuous-domain economic certificate, compare it with an architecture-matched direct neural HJB solve, and retain a failed joint-loss ablation. Neural preference adjustment, dense nonquadratic capital models, actual trace-estimator training, and dynamic fixed-rival best responses extend the same program beyond exactly representable calibrations. Their error accounts remain distinct: a finite-game equilibrium is not a continuous-game theorem, and sampled high-dimensional residuals do

not certify policy payoffs outside their verification region. This separation permits substantive computational progress without changing the economic agenda or treating numerical convenience as an optimality argument.

APPENDIX A: PROOFS

A.1. Proof of Theorem 1

Extend the aggregator Lipschitz-continuously outside its invariant interval for this comparison argument. The extension leaves all policy utilities and the candidate residual unchanged. For a nonnegative number c , define $d_c(t) = Be^{L(T-t)} + cA_L(T-t)$. Then $d_c(T) = B$, $d_c(t) \geq B$, and $-d'_c = Ld_c + c$. Since a spatially constant shift leaves the generator unchanged and f is L -Lipschitz in utility,

$$F[v + d_c] \geq F[v] - d'_c - Ld_c = F[v] + c, \quad F[v - d_c] \leq F[v] + d'_c + Ld_c = F[v] - c.$$

Because $F[v] = r^a[v] - g^a[v] \in [-e - \eta, e]$, $v + d_{e+\eta}$ is a supersolution and $v - d_{e+\eta}$ a subsolution. They dominate the required stopping data on every boundary face. A spatially constant shift also preserves an exactly satisfied homogeneous reflected boundary condition. Comparison proves (13).

Apply the same argument to the fixed-policy operator $F^a[v] = -v_t - \mathcal{H}^a[v]$, with $c = e$, to obtain (12). The triangle inequality gives (14), and policy admissibility gives its nonnegative lower bound. The construction uses $L \geq 0$ and $d_c(t) \geq B$ specifically to cover lateral stopping boundaries; it does not incorrectly discount their errors as if every path stopped only at T . For nonsmooth candidates the same argument applies when their residual inequalities are verified in the viscosity sense. The theorem's differential implementation, however, requires the stated smooth candidate. Q.E.D.

A.2. Exact policy improvement

Suppose $v = V^a$ solves the fixed-policy problem and a feasible $\tilde{a}$ satisfies $\mathcal{H}^{\tilde{a}}[v] \geq \mathcal{H}^a[v]$ pointwise. Then $F^{\tilde{a}}[v] \leq 0$ with the same boundary payoff. Policy comparison gives $V^{\tilde{a}} \geq V^a$. This is a policy-space statement. A single gradient step on a domain-averaged actor loss does not automatically meet its pointwise premise.

The measured action gap and policy-evaluation errors are retained precisely because approximate neural improvement can violate that premise.

A.3. Proof of Theorem 2

For a fixed state and action, let z and $\tilde{z}$ solve (15) for w and $\tilde{w}$. Strict monotonicity of $z - hf(z)$ gives

$$|z - \tilde{z}| \leq \frac{|P_h^a(w - \tilde{w})|}{1 + \lambda h} \leq q_h \|w - \tilde{w}\|_\infty.$$

The same monotonicity implies order preservation. Taking the maximum over actions preserves both order and the Lipschitz constant. Banach's fixed-point argument supplies the two fixed points on the invariant complete value domain.

Now $\|v - v_h^a\| \leq \|v - T_h^a v\| + q_h \|v - v_h^a\|$, which gives the first bound. Also $\|v - T_h v\| \leq \delta + \epsilon$, giving the second. The triangle inequality gives the third. All norms are over every state of the specified finite model. An omitted action changes T_h and must be accounted for as a separate error, not hidden in δ . *Q.E.D.*

A.4. Proof of Theorem 3

First, (16) gives

$$\|v_h - v_h^*\|_\infty \leq \frac{\delta_h + \epsilon_h}{1 - q_h} = \frac{1 + \lambda h}{\lambda} \frac{\delta_h + \epsilon_h}{h} \longrightarrow 0.$$

It remains to identify the limit of the exact monotone scheme. Stability makes the upper and lower half-relaxed limits finite. At a strict local maximum of the upper limit minus a smooth test function, select nearby maximizing grid points. Monotonicity compares the neighboring scheme values to that test function plus the vanishing contact offset. Local consistency then gives the viscosity subsolution inequality. The corresponding minima give the supersolution inequality for the lower limit. Consistent boundary data and the assumed barriers give the required boundary inequalities. Comparison orders the upper limit below the lower limit. The reverse order follows by definition, so the two limits coincide with the unique continuous viscosity solution. The vanishing distance between v_h and v_h^* transfers that convergence to the neural arrays. The discrete policy-value conclusion follows from the

third bound. Interpreting a discrete policy as a continuous-time implementation additionally requires convergence of its controlled interpolation and payoff; the theorem does not infer this merely from convergence of its state arrays. This is the usual monotone-scheme selection argument, with the neural approximation error made explicit; see [Barles and Souganidis \(1991\)](#), [Crandall et al. \(1992\)](#). *Q.E.D.*

A.5. Finite-horizon Bellman error accumulation

Let T_n and T_n^a be the time- n Bellman and fixed-policy maps, with Lipschitz constants q_n (which may exceed one). For arrays v_n , define $\delta_n = \|v_n - T_n^a v_{n+1}\|$ and $\epsilon_n = \|T_n v_{n+1} - T_n^a v_{n+1}\|$. If the terminal error is B_N , backward induction gives

$$\|v_n - v_n^*\| \leq \left(\prod_{j=n}^{N-1} q_j \right) B_N + \sum_{k=n}^{N-1} \left(\prod_{j=n}^{k-1} q_j \right) (\delta_k + \epsilon_k).$$

The policy-evaluation bound replaces $\delta_k + \epsilon_k$ by δ_k . Their sum bounds regret. Empty products equal one. For the implicit Lipschitz recursive step, $q_n \leq (1 - hL)^{-1}$; for additive exponential discount, $q_n = e^{-\rho h}$. Thus a per-step error $o(h)$ and consistent terminal data suffice on a fixed finite horizon, without asserting a stationary contraction in an inapplicable utility regime.

A.6. Derivation of the temporal-self condition

A self at (t, x) replaces a^* by a fixed feasible local action a for duration ϵ and then returns to a^* . For each exponential component, the deviation value is

$$\mathbb{E} \left[\int_t^{t+\epsilon} e^{-\rho_j(s-t)} u(a) ds + e^{-\rho_j \epsilon} W_j(t + \epsilon, S_{t+\epsilon}^a) \right].$$

Itô expansion gives its first-order increment relative to $W_j(t, x)$ as $\epsilon[u(a) + (W_j)_t + \mathcal{L}^a W_j - \rho_j W_j] + o(\epsilon)$. Weighting the two components and subtracting their evaluation equations under a^* leaves

$$\epsilon\{u(a) - u(a^*) + (\mathcal{L}^a - \mathcal{L}^{a^*})V\} + o(\epsilon).$$

Nonpositive first-order gain for every feasible action is exactly (43). This derivation assumes the smoothness and local integrability needed for the expansion; it is an

equilibrium deviation condition, not an assertion of optimal commitment over a finite interval. The homothetic derivation follows by integrating $e^{-[\rho_j - b(r-m)]s}$ and applying the consumption first-order condition. *Q.E.D.*

APPENDIX B: REPLICATION AND SCOPE OF NUMERICAL EVIDENCE

The revision directory contains the executable block core, the experiment program, the regression tests, raw numerical outputs, and a response-to-referee document. The reviewed manuscript, supplement, and historical comments are preserved byte-for-byte in the archive, and the review branch remains unchanged. The new main manuscript is authoritative for the revised equations. The archive is not compiled into the revised article and its unsupported numerical statements are not reused as evidence.

The inherited R2 execution contains 29 recorded experiments and eight regression tests, all retained and rerun. The R6 record adds nine primary stopped-consumption runs, six architecture/update sensitivities, six full-domain certificates, six principal NDU neural runs, sixteen NDU reference sensitivities, twenty-four dense-capital primary fits, six stochastic-trace fits, twelve coverage follow-ups, and six principal dynamic-game solves. Pilot failures and coarse-game counterchecks are retained separately rather than counted as successful primary runs. The original referee diagnostics continue to concern their pinned original sources.

The initial development summary and the corresponding script are preserved separately. Inspection found a sign error in the *reported* quadratic residual formula even though the Lyapunov solutions agreed with Riccati values. The formula was corrected, its acceptance test strengthened to include that residual, and the complete experiment suite rerun. The wealth stopping boundary was also imposed explicitly at all NDU time levels before the final run. The earlier results are marked superseded diagnostics rather than silently relabelled as final evidence.

The R6 directory contains complete source, generated tables, model and numerical arrays, optimizer histories, certificate covers, tests, and a provenance manifest. The replication workflow materializes the revision source, independently reruns its computations, and commits the resulting evidence only to the designated R6

candidate branch; an exact-source snapshot and raw-result hashes identify each execution. Local execution, remote replication, and independent referee acceptance remain different facts. No historical GPU benchmark, unexecuted deep-BSDE comparison, unrestricted neural Cournot equilibrium, or empirical identification result is inferred from the computations reported here.

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